Amogh Ukkadgatri

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Electronics & Communication Engineer · Embedded Systems · RF & Signal Processing

SUMMARY

Hardware engineer in defence R&D at SSS Defence, working on acoustic counter-UAS: microphone arrays, embedded capture firmware, the test rigs that qualify them, and packet, data and time synchronization across multiple arrays. Designs boards end to end, from 4-layer KiCad layout to encrypted LoRa firmware. Suited to regulated, high-reliability defence and aerospace work.

EXPERIENCE

SSS Defence

Sep 2025 – Present

Bengaluru, India (on-site) · DeepTech R&D — Acoustic counter-UAS

Hardware Engineer Sep 2026 – Present

- Part of the team building packet transport, data alignment and time synchronization across multiple microphone arrays, so captures from separate arrays line up for joint processing.
- Continuing to own the acoustic C-UAS sensing hardware, from microphone arrays and capture firmware to the rigs that qualify them.

R&D Intern Sep 2025 – Sep 2026

- Recruited into the acoustic C-UAS program on the strength of a final-year sound source localization project, to field that array and DOA work.
- Built the microphone arrays for the first acoustic C-UAS prototype on Raspberry Pi 4 — array geometry, sensor selection and capture across multiple communication protocols — and adapted a triangulation-based localization approach into the pipeline so the team's classical and physics-based algorithms could run on live multi-channel data.
- Built and validated the test rigs that qualify those mic arrays in hardware (see Projects): repeatable capture, channel-alignment and bearing-accuracy checks against known source positions, with DHCP, static IP and subnet masking across prototypes for reliable Ethernet transfer.

SELECTED PROJECTS

3D Sound Source Localization using a Mic Array Sep 2025 – Mar 2026

Real-time audio sensing system that detects sound sources and resolves their direction in three dimensions using classical DOA methods over a custom-built microphone array — array geometry and calibration, multi-channel synchronous capture and a processing chain that holds bearing accuracy on live signals in noisy rooms. The work this internship was offered on.

Real-time 3D Radar Tracking & Classification System Sep – Nov 2024

Radar system that tracks moving objects in three dimensions and classifies them by their distinct return signatures — range–Doppler processing, track association across frames and a feature-based classifier separating object types in real time.

Theoros — Encrypted GPS Logger with LoRa Telemetry Oct 2026

Designed a USB-powered ESP32-S3 GPS tracker end to end: a 4-layer KiCad board built for machine assembly, with a u-blox SAM-M10Q GNSS module, RFM95W 866 MHz LoRa radio, microSD and OLED, and impedance-checked RF and USB traces. Wrote the tracker and receiver firmware: LoRa telemetry and SD logs encrypted and authenticated with AES-128-CCM, plus WiFi NTP time and OTA. Ready for fabrication.

Mic-Array Validation Test Rig — ESP32 I2S PDM / TDM 2025 – 26

Built a complete test rig single-handedly to validate custom mic arrays over ESP32 I2S PDM capture, solving the classic ESP32's single-RX-channel limit by moving to TDM.

SKILLS

Languages & tools	C, Python, MATLAB, Verilog (basics), KiCad, Audacity, Linux
Hardware & embedded	ESP32 / ESP32-S3, Raspberry Pi, Arduino, 8051, I2S / PDM / TDM audio, LoRa, GNSS, 4-layer PCB design, sensors, op-amps, MOSFETs, transistors, soldering and firmware flashing
Systems & RF	Link-budget & detection analysis, antenna/array design, full-wave EM simulation, digital back-end architecture, multi-board layout, signal processing
Networking	DHCP, static IP, subnet masking, Ethernet data transfer

EDUCATION

B.E., Electronics & Communication — KLE Institute of Technology · **CGPA 8.84/10** 2022 – 2026

Senior Secondary (XII), Science — Chetan PU College, Karnataka State Board · **81.00%** 2022

Secondary (X) — Oriental Public School, State Board · **96.16%** 2020

LEADERSHIP & ACTIVITIES

Chair, KLEIT–IEEE MTT-S Chapter (2024–25) · Vice-Chair, KLEIT–IEEE Student Branch (2023–24) · Coordinator, Technical & Cultural Fest · Class Representative (2023–24) · University Cultural Group (Theatre)